FICTION / PROBABILITY / A MATHEMATICAL INVESTIGATION

The Paradox of Absolute Luck

Can probability favor everyone?

An original mathematical essay inspired by Hardcore Leveling Warrior. *The scenes and numerical experiments below are invented thought experiments, not measured events from the webtoon.*

The duel with two guaranteed winners

Imagine an arena with only one rule: **exactly one person leaves victorious**. A mysterious power whispers to Fighter A, "You have a 99% chance of winning." Fighter B enters with the same power—and hears precisely the same promise.

For a moment, both promises sound reasonable. Each fighter is extraordinarily lucky. But the mathematics has a problem. Let W_A mean "A wins" and W_B mean "B wins." Because one victory excludes the other,

$$P(W_A) + P(W_B) \leq 1.$$

Two simultaneous 99% promises instead demand $0.99 + 0.99 = 1.98$. It is tempting to say luck has broken probability. The more interesting answer is that our *description* of luck is incomplete. **Ninety-nine percent against whom? In which contest?**

One possible model gives each competitor a positive strength w. In a two-person contest, the chance of winning is the player's strength divided by the two strengths combined:

$$P(A \text{ wins against } B) = \frac{w_A}{w_A + w_B}.$$

Give each lucky fighter strength 99 and an ordinary opponent strength 1. Either lucky fighter beats the ordinary opponent 99% of the time. Against each other, however, they each win 50% of the time. No guarantee was broken: the two 99% statements belonged to *different matchups*. This is an illustrative use of the established Bradley–Terry/Luce family of choice models, not a law of supernatural luck [1,2].

Nor must the duel be 50–50. Equal records against an ordinary opponent do not, by themselves, establish equal chances against each other. The real lesson is that a winning percentage cannot float free of its opponent and setting.

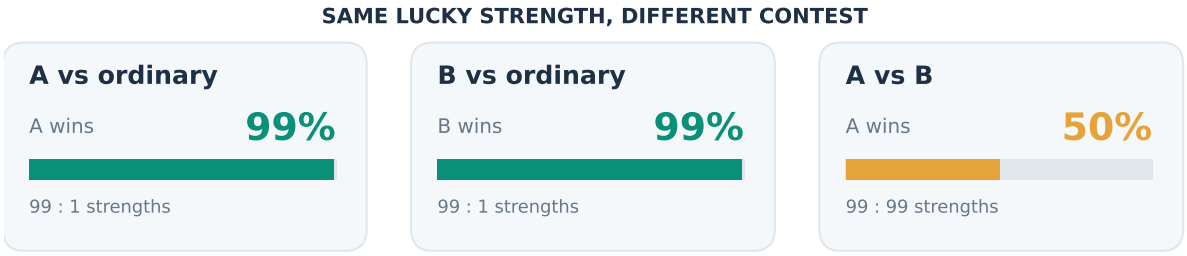


Figure 1: Two 99% outcomes against different opponents do not imply two 99% winning chances in the same duel.

Three coins, and the trap of independence

Now replace the duel with coins. Suppose each of three coins succeeds with probability 99.5% **while all three are being used together**. Can the probability that all three succeed drop to 60%?

Call failure of coin i the event F_i. For at least one coin to fail, one of F_1, F_2, F_3 must occur. Even when failures are dependent, the union bound gives

$$P(F_1 \cup F_2 \cup F_3) \leq 0.005 + 0.005 + 0.005 = 0.015.$$

So the probability that *all three succeed* is at least $1 - 0.015 = 98.5\%$. A collective 60% success rate cannot coexist with all three **same-context** marginals remaining at 99.5% [3]. It would require the individual rates to change, or the original statement to refer to another situation—for example, activating a coin alone.

The phrase “almost certain” is not enough. We need to know whether the probabilities describe isolated coins, a simultaneous batch, or a particular relationship among outcomes.

Two hundred coins and no possible perfect run

Now comes the seemingly impossible twist. Increase the number of coins to **200**. Imagine 200 equally likely possible worlds. In world 1, only coin 1 fails. In world 2, only coin 2 fails. Continue this pattern until world 200, where only coin 200 fails.

Choose a world uniformly at random. Any particular coin fails in exactly one world out of 200. Therefore,

$$P(\text{coin } i \text{ succeeds}) = \frac{199}{200} = 99.5\%.$$

But *every* world contains a failure. There is not a single world in which all 200 coins succeed:

$$\boxed{P(\text{all 200 succeed}) = 0.}$$

We did not lower the reliability of any coin. We changed **how their failures fit together**. In the three-coin example, there were too few small failure probabilities to cover every possibility. With 200 coins, 200 separate half-percent failure events can exactly cover the entire probability space.

Independence would tell a completely different story. If the 200 coins instead succeeded independently with probability 0.995 each, then

$$P(\text{all succeed}) = (0.995)^{200} \approx 36.70\%.$$

Same 99.5% for each coin. One model gives 0% joint success; another gives approximately 36.70%. A third mathematically valid shared-parameter model gives approximately 49.87%. **Individual luck alone cannot determine collective luck.**

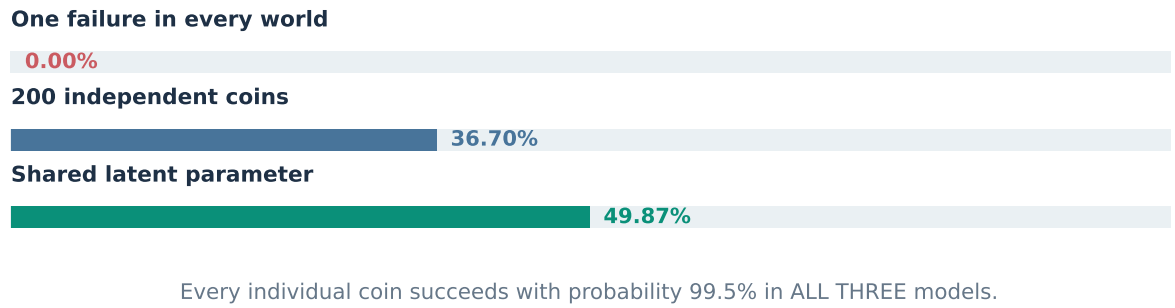


Figure 2: Three exact joint-success probabilities under different stipulated dependence mechanisms; all 200 individual marginals equal 99.5%.

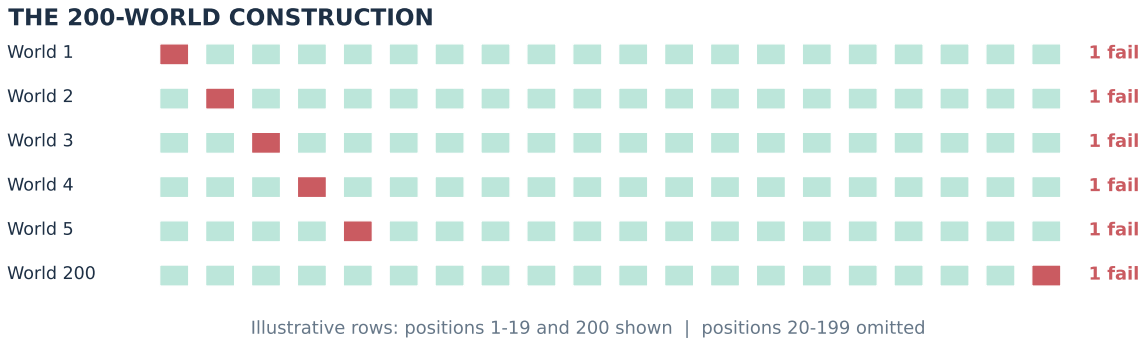


Figure 3: Only representative rows and coin positions are pictured; the construction has 200 complete equally likely worlds, with one failure per world.

The coin that cannot be added

Our zero-success construction is symmetric: swapping coin labels changes nothing. Mathematicians call this *exchangeability*. Surprisingly, the exact 200-coin construction cannot be extended to **201 exchangeable coins** while keeping the original joint law for the first 200 [4].

Here is an intuitive way to see why. In any exchangeable 201-coin extension, *every choice of 200 coins* would have to contain exactly one failure, with certainty. If all 201 coins contained exactly one failure, omitting the failed coin would leave none. If they contained two failures, omitting a successful coin would leave two. Zero failures or three or more are no better. No arrangement works for every possible omission.

This does **not** mean the universe forbids a 201st coin. It means the exact probability model cannot preserve all of those symmetry requirements at once. Finite and infinitely extendible

probabilistic models are not interchangeable; that distinction is part of established exchangeability theory [4,5].

The information missing from every pair

Suppose a researcher watches three coins and finds that each succeeds 70% of the time, and any chosen pair succeeds together 50% of the time. Surely that tells us the chance all three succeed?

Not quite. Construct two models, L and H. Both obey those **identical** single-coin and pairwise statistics. Yet model L gives a 30% chance of all three succeeding, while model H gives 40%.

Nothing is wrong with the measurements. They simply do not reveal the three-way dependence. If observations are restricted to a fresh single coin or pair on each trial, these two models generate exactly the same observable data. To distinguish them, one must sometimes see the **whole triple from the same trial**.

We often ask whether two things are correlated. Absolute Luck suggests a subtler question: *What relationships appear only when three or more things happen together?*

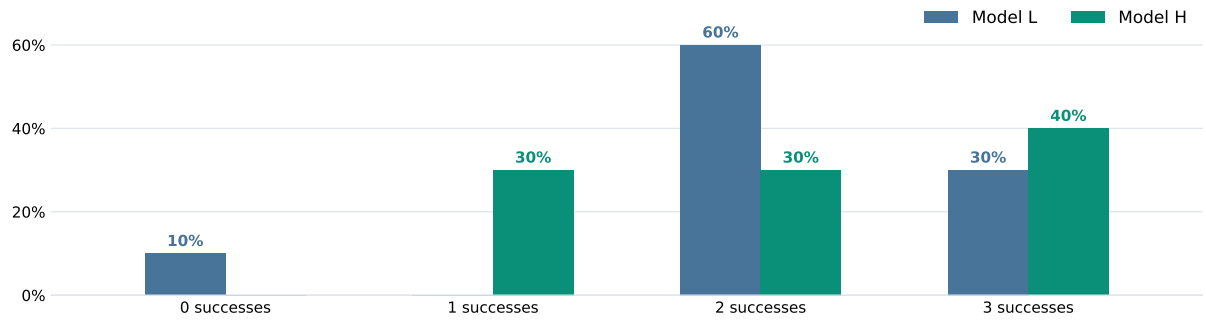


Figure 4: Identical single-coin and pairwise probabilities can conceal different distributions of the number of successes.

The best use of luck depends on the rules

Imagine a limited probability boost of 49.5 percentage points, shared among three coins that each start at 50%. We can distribute it equally, making each coin 66.5% reliable, or concentrate it on the first coin, giving probabilities 99.5%, 50%, and 50%.

If outcomes are independent, equal sharing wins when the goal is for **all three** to succeed: $0.665^3 \approx 29.41\%$, compared with $0.995(0.5)(0.5) = 24.875\%$.

Now change the dependence rule but **hold that new rule fixed across both strategies**. Draw one uniform random number U between zero and one. Coin 1 succeeds when $U \geq 1 - p_1$; coins 2 and 3 succeed when $U \leq p_2$ and $U \leq p_3$. All individual success rates remain correct. Yet the equal strategy now yields 33% collective success, while concentration yields 49.5%.

The ranking has reversed. This comparison concerns these two strategies under an illustrative mechanism; it does **not** prove concentration is globally optimal. It shows why an instruction as simple as “spread luck evenly” needs a dependence assumption.

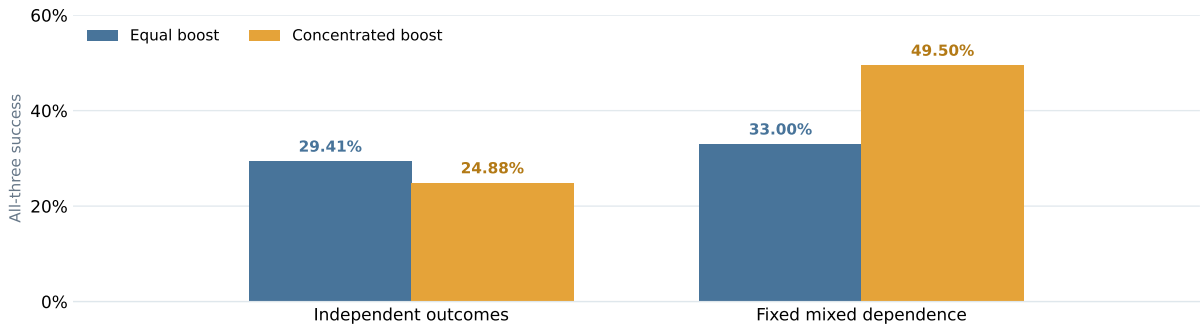


Figure 5: Two allocation strategies reverse their ordering when the joint dependence mechanism changes. Both bars within each setting use the same coupling.

What a computer can—and cannot—prove

Our companion mathematical study ran seeded simulations with 120,000 trials per principal scenario and checked them against exact formulas. For the 200-coin cases, the simulated all-success frequencies were 0%, 36.57%, and 49.75% under three different stipulated mechanisms, close to their theoretical values. The simulations reproduce models we designed; they are **not observations from the comic**, nor are they substitutes for proofs.

The fictional inspiration is *Hardcore Leveling Warrior*, credited to Sehoon Kim [6]. We have not independently authenticated precise activation percentages from its original panels. Every percentage in this essay belongs to a clearly stated mathematical thought experiment.

The answer hiding inside the paradox

Can probability favor everyone? It can favor two people against the same ordinary opponent; it cannot assign both a 99% win probability in the very same single-winner duel. It can make every one of 200 coins almost certain to succeed while making their perfect collective success impossible. It can even conceal the most important relationship from every pairwise observation.

Absolute luck is not a complete probability statement. To give it meaning, we must say *which event, which opponent, which group, and which dependence structure*. The paradox does not reveal a crack in mathematics. It reveals the assumptions hidden inside a promise of certainty.

Sources and further reading

[1] Bradley & Terry (1952), “Rank Analysis of Incomplete Block Designs: I.” *Biometrika* 39, 324–345. DOI.

[2] Luce (2008), “Luce’s choice axiom.” *Scholarpedia* 3(12):8077. DOI.

[3] Hoppe & Nediak (2008), “Fréchet optimal bounds on the probability of a union with supplementary information.” *Statistics & Probability Letters* 78, 311–319. DOI.

[4] Konstantopoulos & Yuan (2019), “On the extendibility of finitely exchangeable probability measures.” *Transactions of the AMS* 371, 7067–7092. DOI.

[5] Kirsch (2019), “An elementary proof of de Finetti’s theorem.” *Statistics & Probability Letters* 151, 84–88. DOI.

[6] Kim, *Hardcore Leveling Warrior*. Official WEBTOON series.